LAPORAN TUGAS: MVC SEDERHANA

Tabel Genre dan Author - Sistem Informasi Penjualan Buku (Book Sales)

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1. Tujuan

Membuat sistem MVC sederhana di Laravel untuk tabel Genre dan Author pada Brief Book Sales. Data disimpan sebagai array di Model (masing-masing 5 data), diambil oleh Controller melalui Route, lalu ditampilkan oleh View.

2. Komponen MVC yang Dibuat

Komponen	Genre	Author
Route	routes/web.php: /genres, /genres/{id}	routes/web.php: /authors, /authors/{id}
Controller	GenreController.php	AuthorController.php
Model	Genre.php (5 data)	Author.php (5 data)
View	genres/index, genres/show	authors/index, authors/show

Alur: Route menerima URL, lalu memanggil Controller. Controller mengambil data dari Model dan mengirimkannya ke View untuk ditampilkan di browser.

3. Kode Model

3.1 Genre.php (kolom: id, name, description)

```
<?php

namespace App\Models;

class Genre
{
    // Data array statis (struktur sesuai tabel genres: id, name, description)
    public static function all(): array
    {
        return [
            ['id' => 1, 'name' => 'Fiksi',       'description' => 'Cerita rekaan yang lahir dari imajinasi
penulis.'],
            ['id' => 2, 'name' => 'Non-Fiksi',   'description' => 'Buku berbasis fakta, pengetahuan, dan
kejadian nyata.'],
            ['id' => 3, 'name' => 'Fantasi',     'description' => 'Cerita dengan dunia sihir, makhluk mitos,
dan petualangan.'],
            ['id' => 4, 'name' => 'Teknologi',   'description' => 'Buku seputar pemrograman, komputer, dan
inovasi digital.'],
            ['id' => 5, 'name' => 'Sejarah',     'description' => 'Catatan dan analisis peristiwa masa
lalu.'],
        ];
    }

    public static function find(int $id): ?array
    {
```

```
        foreach (self::all() as $genre) {
            if ($genre['id'] === $id) {
                return $genre;
            }
        }
        return null;
    }
}
```

3.2 Author.php (kolom: id, name, photo, bio)

```
<?php

namespace App\Models;

class Author
{
    // Data array statis (struktur sesuai tabel authors: id, name, photo, bio)
    public static function all(): array
    {
        return [
            ['id' => 1, 'name' => 'Andrea Hirata',          'photo' => 'andrea-hirata.jpg',  'bio' =>
'Novelis Indonesia, penulis tetralogi Laskar Pelangi.'],
            ['id' => 2, 'name' => 'Pramoedya Ananta Toer',  'photo' => 'pramoedya.jpg',      'bio' =>
'Sastrawan Indonesia, penulis Tetralogi Buru.'],
            ['id' => 3, 'name' => 'Tere Liye',              'photo' => 'tere-liye.jpg',      'bio' =>
'Penulis produktif novel populer Indonesia.'],
            ['id' => 4, 'name' => 'Dee Lestari',            'photo' => 'dee-lestari.jpg',    'bio' =>
'Penulis dan penyanyi, pencipta seri Supernova.'],
            ['id' => 5, 'name' => 'Robert C. Martin',       'photo' => 'robert-martin.jpg',  'bio' =>
'Software engineer, penulis Clean Code.'],
        ];
    }

    public static function find(int $id): ?array
    {
        foreach (self::all() as $author) {
            if ($author['id'] === $id) {
                return $author;
            }
        }
        return null;
    }
}
```

4. Route dan Controller

```
<?php

use App\Http\Controllers\AuthorController;
use App\Http\Controllers\GenreController;
use Illuminate\Support\Facades\Route;

Route::get('/', fn () => redirect('/genres'));

Route::get('/genres', [GenreController::class, 'index'])->name('genres.index');
Route::get('/genres/{id}', [GenreController::class, 'show'])->name('genres.show');

Route::get('/authors', [AuthorController::class, 'index'])->name('authors.index');
Route::get('/authors/{id}', [AuthorController::class, 'show'])->name('authors.show');
```

```
<?php

namespace App\Http\Controllers;

use App\Models\Genre;

class GenreController extends Controller
```

```
{
    public function index()
    {
        return view('genres.index', ['genres' => Genre::all()]);
    }

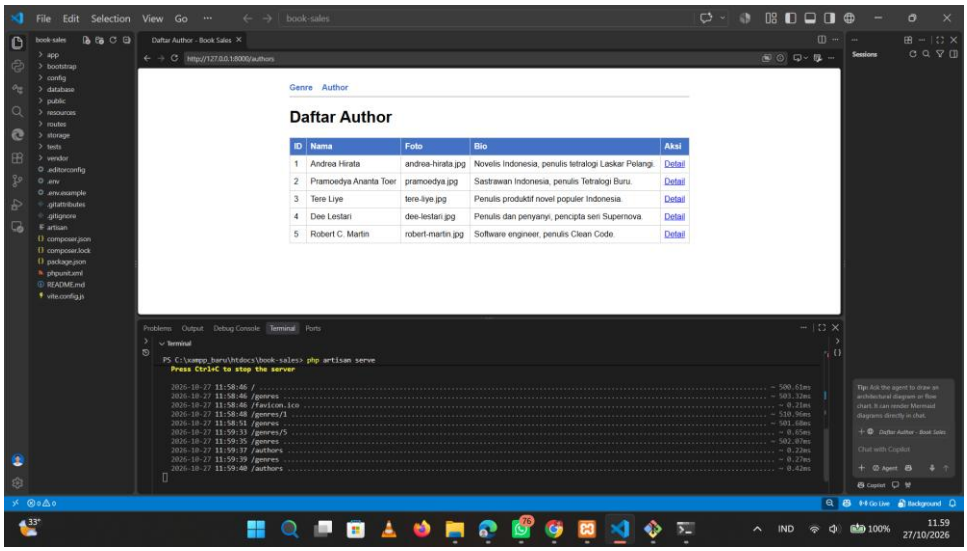
    public function show(int $id)
    {
        $genre = Genre::find($id);
        abort_if(is_null($genre), 404);

        return view('genres.show', ['genre' => $genre]);
    }
}
```

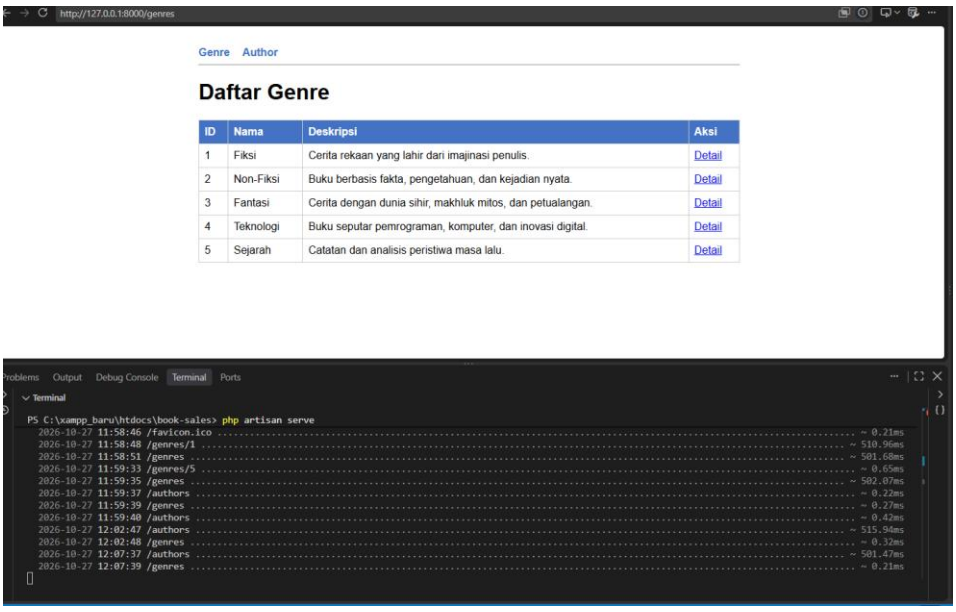
AuthorController memiliki struktur yang sama dan memakai Model Author serta view authors.*.

5. Hasil

Aplikasi dijalankan dengan php artisan serve. Halaman /authors dan /genres masing-masing menampilkan tabel 5 data.



Gambar 1. Halaman /authors



Gambar 2. Halaman /genres

6. GitHub

Link repository: https://github.com/muhammadaditia433/Framework-Laravel.git

File Model yang dikumpulkan: app/Models/Genre.php dan app/Models/Author.php.